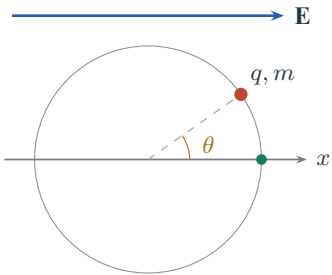
M2-081 约束只要求切向合力为零



图示 $qE_0 > 0$，绿色点稳定；法向电力由约束力平衡。

$$U = -qE_0 R\cos\theta$$

$qE_0 > 0$ ： $\theta = 0$稳定

$$\omega = \sqrt{qE_0/(mR)}$$